



Sujoy Mondal

Senior Full-Stack Engineer | GenAI Systems Architect

PROFESSIONAL SUMMARY

Senior Full-Stack Engineer with 15+ years of experience designing, building and operating production software systems, with recent focus on GenAI application architecture — multi-provider LLM orchestration and retrieval-augmented generation.

Over the last 10+ years as Co-Founder & CTO, served as sole engineering owner of a production AI SaaS platform powering museum and cultural visitor experiences across Europe. Designed and built a custom multi-provider LLM orchestration layer from scratch, and a retrieval-augmented architecture (pgvector, LangChain.js, LangGraph.js) to address the context-window and token-cost limits reached in production.

Specializes in solving production engineering problems through pragmatic architectural decisions under real-world operational constraints.

PROFESSIONAL EXPERIENCE

My App Zone SRL (Neareo), Belgium

Co-Founder & Chief Technology Officer · Dec 2015 – Present

Multi-Model AI Orchestration

- Architected and implemented a **provider-agnostic LLM orchestration layer** from scratch in Node.js/TypeScript — framework-free (pre-LangChain) — supporting OpenAI, Claude, Gemini, DeepSeek, Kimi K2 and fal.ai.
- Designed a **resilient execution pipeline** using tool-call chaining, automatic JSON repair and conditional follow-up model invocation to improve structured-output reliability across providers.
- Unified stateful (OpenAI Assistants) and stateless providers behind a common **orchestration abstraction**, maintaining Redis-backed conversation state, rolling summarization and provider-independent fallback handling.

Retrieval-Augmented Architecture

- Encountered token-bloat and context-window limits in the framework-free orchestration layer (raw context injection requiring rolling summarization every 8 turns).
- Designed a **retrieval-augmented architecture** — introducing LangChain.js, LangGraph.js and pgvector on top of the otherwise framework-free stack — to replace raw context injection with semantic retrieval, improving response quality while reducing token consumption.

Voice & Multimodal AI

- Extended the platform beyond text: the LLM generates narration and image plans, after which the orchestrator fans out parallel voice-synthesis and image-generation requests before aggregating results for delivery.
- Delivered runtime-controlled voice synthesis (Gemini) with dynamic intonation, speaking style and pause insertion for natural museum narration, alongside image generation (fal.ai) — coordinated multimodal output with conditional re-invocation when a stage requires further model calls.

CONTACT

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CAREER HIGHLIGHTS

Co-Founder

Startup · 10+ Years

15+ Years

Production Engineering

10+ Years

Solo Engineering Owner

TECHNICAL STACK

GenAI & LLM

Multi-Model AI Orchestration · RAG · Tool-Calling · Embeddings · LangChain.js · LangGraph.js

Multimodal AI

Voice / TTS (intonation, style, pause) · Image Gen · Text

Backend

Node.js · Express · TypeScript · Jest

Databases

PostgreSQL · pgvector · MySQL · Redis · Supabase

Frontend

React · JavaScript · TypeScript · Tailwind CSS

AI Providers

OpenAI · Claude · Gemini · DeepSeek · Kimi K2 · fal.ai

Infrastructure

Docker · Git · Render · Vercel · DigitalOcean

Runtime

IndexedDB · Service Workers · WebSockets

Payments

Stripe · PayPal

OPEN TO

✓ GenAI Full-Stack Engineer

✓ AI Systems Architect

✓ Founding Engineer

✓ **Availability: Immediate**

AI SaaS Platform & Scale

- Architected and operated a production AI SaaS platform powering museum and cultural visitor experiences across Europe, serving users from 35+ countries.
- Designed the platform’s **no-code builder**, enabling cultural institutions to compose AI-driven, multimodal visitor experiences without engineering effort.

Offline-First Runtime & Analytics

- Designed an **offline-first browser execution runtime** using IndexedDB, Service Workers, adaptive media pre-caching and checkpoint/restore mechanisms to maintain application continuity without reliable connectivity.
- Redesigned **analytics processing** from SQL-heavy aggregation to precomputed, IndexedDB-backed architectures, reducing runtime computation under production workloads.
- Reconstructed **dynamic AI conversation journeys** from production interaction data, enabling per-visitor analytics over non-deterministic multi-turn LLM flows.

Real-Time Sync & Payments

- Designed **distributed synchronization mechanisms** using WebSockets to coordinate synchronized playback across kiosks and multiple client devices.
- Integrated payment and ticketing infrastructure across Stripe, PayPal and kiosk payment providers within the production platform.

Node.js

TypeScript

LangChain.js

LangGraph.js

pgvector

PostgreSQL

Redis

MySQL

IndexedDB

WebSockets

Docker

Jest

DigitalOcean

SocialBuzz Advertising, Brunei Darussalam · Senior Programmer & Web Developer (Virtual) · Dec 2013 – Nov 2015

Owned architecture, development and production operations across multiple e-commerce and SaaS platforms, from requirements through deployment and support.

WebSkitters Technology Solutions Pvt. Ltd. · Software Developer · Dec 2012 – Nov 2013

Built complex browser-based applications including interactive product configurators and state-driven customization systems.

ARTIS Technologies Pvt. Ltd. · Senior Developer · Oct 2011 – Nov 2012

Built browser automation workflows and OCR-assisted data extraction pipelines processing hundreds of thousands of records.

Engineering Portfolio & Case Study Platform (2026)

- Designed and built a public engineering platform documenting production architecture, runtime systems and platform-evolution case studies from 15+ years of production engineering.
- Implemented responsive frontend architecture, backend APIs, SEO optimization and Open Graph integration.
- Deployed using Vercel and Render with PostgreSQL-backed persistence and Git-based CI/CD workflows.

React

Vite

Tailwind CSS

Node.js

Express

PostgreSQL

Supabase

Render

Vercel

EDUCATION

Bachelor of Technology (B.Tech.) — Computer Science & Engineering

Dumkal Institute of Engineering & Technology · 2004 – 2008